

Forward Chaining

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Program 1:

File Edit View

```
% Initial knowledge
start :-
    assertz(known(a)),
    assertz(known(b)),
    infer.

% Rule
infer :-
    known(a),
    known(b),
    \+ known(c),
    assertz(known(c)),
    write('Inferred c'),
    nl.
```

```
50 ?- consult(['c:/Users/Lenovo/Desktop/program - 91.pl']).
true.
```

```
50 ?- start.
Inferred c
true.
```

Program 2:

File Edit View

```
% Rule to infer fire
check :-
    known(smoke),
    known(heat),
    assertz(known(fire)),
    write('Fire Alert!'),
    nl.

% Start rule
start :-
    assertz(known(smoke)),
    assertz(known(heat)),
    check.
```

```
51 ?- consult(['c:/Users/Lenovo/Desktop/program - 92.pl']).
```

```
Warning: c:/users/lenovo/desktop/program - 92.pl:10:
```

```
Warning:   Redefined static procedure start/0
```

```
Warning:   Previously defined at c:/users/lenovo/desktop/program - 91.pl:2
true.
```

```
52 ?- start.
```

```
Fire Alert!
```

```
true.
```

Program 3:

File Edit View

```
% Rule to infer pollution
infer_pollution :-
    known(cloudy_water),
    known(bad_smell),
    assertz(known(polluted)),
    write('Water is polluted'),
    nl.

% Start rule
start :-
    assertz(known(cloudy_water)),
    assertz(known(bad_smell)),
    infer_pollution.
```

```
53 ?- consult(['c:/Users/Lenovo/Desktop/program - 93.pl']).
Warning: c:/users/lenovo/desktop/program - 93.pl:10:
Warning:   Redefined static procedure start/0
Warning:   Previously defined at c:/users/lenovo/desktop/program - 92.pl:10
true.

54 ?- start.
Water is polluted
true.
```